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Place-based preferential tax policy and industrial development: Evidence from India's program on industrially backward districts☆



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ABSTRACT

We evaluate a tax-exemption program initiated by the Indian government in 1994 to promote manufacturing in districts designated as industrially backward on the basis of a continuous gradation score that reflected district characteristics in early 1990s. Employing a regression discontinuity design, we find that the program led to a significant increase in firm entry and employment, especially in light manufacturing industries of the better-off backward districts in the short run. However, this was partly driven by spatial displacement of economic activity from neighboring districts that narrowly missed qualifying for the program. Further, we do not find the effects of the program to persist after it ended.

1. Introduction

Place-based policies have been popular in both developed and developing countries. Examples of large scale place-based policies include the federal Empowerment Zone Program in the United States established in 1993, European Union (EU)'s various initiatives supported under its Structural Funds targeted at disadvantaged areas and countries within EU, and China's Special Economic Zones starting in late 1970s, just to name a few (Neumark and Simpson, 2015).

A common goal of place-based policies is to create jobs and spur economic activity by attracting new firms to and/or promoting growth of existing firms in selected areas. The policies usually take one or a combination of the following forms: tax exemptions and subsidies, discretionary grants, special economic zones or industrial parks, and infrastructural support. Place-based policies are often targeted at certain types of industries. For instance, some industrial parks are established to host firms in high-technology manufacturing only.

This paper studies the backward district program of India which was initiated in 1994 and offered new firms a 5-year tax exemption on profits and another 5-year tax reduction afterwards. The program is a typical place-based policy as it was only applicable to 123 industrially backward districts identified out of 360 districts from 14 major states of India. The program was focused only on firms engaged in manufacturing activities.

We attempt to examine both the short-term and long-term impacts of the program with two rounds of economic census data administered in 1998 and 2005, respectively. A key challenge in evaluating the impact of place-based policies lies in the fact that treated areas are often not randomly selected. For a program supporting underdeveloped areas, not only do the treated and untreated areas differ in initial conditions, their development trajectories could also be distinct in the absence of the program. In such cases, untreated areas do not constitute good counterfactuals for quantifying program impacts.

The approach used by the Indian government to identify backward districts, however, offers us a unique opportunity to credibly evaluate the program's impacts. In short, the central government calculated a composite gradation score for each district across 14 major states based on

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several indicators such as the urbanization rate and per capita gross value added in the registered manufacturing sector in the early 1990s. Districts with scores below a specific cutoff point were designated as backward districts qualifying for the tax exemption scheme. This setting allows us to use a (sharp) regression discontinuity (RD) design in identifying the causal effects of the program. Upon checking the distribution of the gradation scores, we show that there is no bunching of districts near the cutoff point. Moreover, non-backward districts with scores right above the cutoff point are statistically indifferent from backward districts right below the cutoff in terms of pre-treatment covariates such as population, labor participation rate, and number of manufacturing workers. Hence, RD design, especially applied to districts near the cutoff point should yield plausible estimates of the program's effects.

Our main findings are as follows. First, the program led to remarkable growth in the manufacturing sector in the better-off backward districts, which account for about one third of all backward districts with gradation scores nearest to the cutoff, over four years after the policy was introduced. The effects largely accrued to light manufacturing industries (with about a 60% increase in both number of firms and employment), while some results suggest a modest increase (about 20%) in heavy manufacturing industries. The results prove to be highly robust when a variety of alternative sample treatments, estimation methods, and geographically overlapping policies are considered. A falsification test also lends credibility to the results.

To understand how much of the growth effect arose from a net increase in new economic activity, we investigate the effects of the program on firm formalization and spatial spillovers generated by the program. The evidence suggests that the program did not have a sizable effect on the share of formal firms or their employment in backward districts as of 1998. We do find some evidence suggesting net negative spatial spillovers, although the spatial displacement was not large enough to offset the direct effects of the program on the better-off backward districts. In sum, the analyses support the notion that the program promoted manufacturing development in the treated areas in the short run and generated overall positive effects even after accounting for negative spillovers to the untreated areas.

We do not find the gains in manufacturing growth to persist. In 2005, six years after the program cut its tax benefits by three fourths and one year after the program ended, the estimated effects for manufacturing industries diminish and lose statistical significance. This indicates that the program failed to generate self-sustained agglomeration economies in the treated areas. Thus, the economic returns to the program seem to be rather limited.

This paper contributes to a vast literature on place-based policies in a few ways.[1] First, studies on the effects of economic/enterprise zones in creating jobs yield mixed results. For instance, Neumark and Kolko (2010) find the California Enterprise Zones program to have no significant effects in generating employment, whereas Freedman (2013) finds positive effects for the Texas Enterprise Zones program. Hanson (2009) and Busso et al. (2013) also produce an opposite set of findings with respect to the Federal Empowerment Zones program. Our own findings suggest that the success of a place-based policy may depend on the time frame within which it is examined and the specific interventions it adopts. Mere time-bounded tax exemption may attract firms and contribute to increased employment in the short run, but it is less effective in empowering local communities to generate sustainable economic development in the longer term. Studies that find persistent effects of a temporary place-based policy stress the importance of accumulated capital or agglomeration economies resulting from the policy (Ehrlich and Seidel, 2018; Kline and Moretti, 2014), which may help explain the inefficacy of the program in longer term.

Second, work on the spatial spillovers of place-based policies is relatively sparse. Givord et al. (2013) found strong displacement effects of the French Zones Franches Urbaines (ZFUs) program on the nearby non-ZFU areas, which were of comparable magnitude to the positive effects the program generated inside the ZFUs. However, the results for the US programs are quite mixed (e.g. Ham et al., 2011; Hanson and Rohlin, 2013). In our case, we find net negative spatial spillovers for untreated areas which were geographically and economically close to treated areas, although the magnitude is not large enough to completely offset the positive effects of the program.

This paper also extends a growing literature on place-based policies in developing countries, which can be categorized into two types based on their primary goals. One involves programs to boost economic growth in leading areas such as China's special economic zones, originally established in coastal areas. The other aims to reduce geographical disparities and targets policy interventions at underdeveloped areas. The latter, to which the backward district program belongs to, sometimes bear more political than economic significance.

In an evaluation of China's experiment with special economic zones, Wang (2013) found that they increased the level of FDI and exports as well as average wages of workers in the hosting municipalities and generated a moderate displacement effect in adjacent municipalities. Shenoy (2018) shows that a comprehensive package including infrastructure investment, subsidy and tax exemption promoted economic development in the new state of Uttarakhand, India in early 2000s. Unlike these policies, the backward district program offered only tax exemption to new firms in selected areas.

More similar to the backward district program is India's New Industrial Policy that Chaurey (2017) studies, which offered tax exemptions and capital subsidies for firms in two states, Uttarakhand and Himachal Pradesh, since 2003. Applying a difference-in-differences approach, the author showed that the policy resulted in large increases in outcomes such as employment, number of firms, and total output in treatment states relative to control states. In addition, there is no evidence for any spatial displacement of firms.

However, the two policies differ in a few important aspects. For instance, the backward district program had nationwide coverage with treatment implemented at district level and only new firms eligible for the program. These features enable us to implement a distinctive identification strategy and probably contribute to our different conclusions. In particular, while we obtain similar effects in the short run, these do not persist after the program ended. Moreover, we confirm that attention should be paid to spatial displacement in designing place-based policies in the future.

This paper is also connected to the literature on the location and growth of firms in response to local taxation. The available evidence is somewhat mixed. Some studies, for example, Rathelot and Sillard (2008) find a weak response of firms' location choice to higher taxes, while others, for example, Bartik (1991) and Guimaraes et al. (2004), find a negative relationship. After correcting for potential endogeneity issues, Duranton et al. (2011) find that local taxation has a negative impact on firm employment but no effect on firm entry in the United Kingdom. Our work complements this literature by examining the entry and formalization decisions of industrial firms in response to a large-scale preferential tax policy.

2. Policy background

The spatial pattern of economic activity in India has been characterized by a concentration of industrial development around large cities and development skewed towards a few states. This concentration has grown with time as shown by studies such as that of Ghosh et al. (1998) which reported a rise in the coefficient of variation of per-capita state domestic product from 25% in 1950–51 to 35% in 1993–94. This has occurred despite balanced regional development being a key policy objective of the Indian government since the formulation of its second five-year plan (1956–61).

[1] See Neumark and Simpson (2015) and Duranton and Venables (2018) for detailed reviews of the literature.

India's approach to balanced regional development has evolved from heavy public sector involvement and direct central intervention prior to 1980 to a more market-oriented local approach since 1991 (Planning Commission, 1981; Singhi, 2012). The Industrial Policy Statement of 1991 announced major changes to the government's role, reducing industrial licensing, relaxing industrial location policy, and allowing entry of large enterprises into industries earlier "reserved" for small-scale firms (e.g. Martin et al., 2017; MCI, 1991). Despite a shift towards greater market-orinetation in the post-reform era, a study conducted by the National Institute of Public Finance and Policy in 1987 suggested enlargement of central government tax incentives in the Income Tax Act to encourage entrepreneurship and industrial dispersal. In response, the Finance Act of 1993 introduced a tax holiday scheme for new industrial undertakings[2] located in the so-called "backward" states and union territories, which clustered in the north-eastern and north-western parts of India.

Immediately after the introduction of the 1993 Act, the government received many requests for similar fiscal support to industrially-challenged districts in the remaining 14 non-backward states. A study group was thus constituted under the Ministry of Finance to develop guidelines for identifying backward districts. Specifically, it constructed a composite score based on eight financial, infrastructural, and industrial indicators to approximate a district's degree of industrial development. The individual score on each indicator for each district was calculated as a percentage relative to India's nationwide average using various data sources including the 1991 Census. The overall score is the weighted sum of the eight individual scores with weights equal to 1, 2 or 3 (see Table A1 for more details). We refer to the overall score as "gradation score" hereafter in that it was published in the "All India Gradation List" created through the Finance Act of 1994 as Appendix III of the Income Tax Act.

According to the guidelines, districts that failed to score above 500 were granted backward status, which qualified them for the preferential tax treatment enacted in the Finance Act of 1994. Out of the total 360 districts from the 14 states, 123 were designated industrially backward districts, all of which had gradation scores below 500 except for three districts that were tagged backward because they were either a "no industry" district or an inaccessible hill area district, or did not have a "railhead" as of April 1st, 1994.

Table 1 (left panel) presents numbers of backward and non-backward districts by state. Eleven out of 14 states have both backward and non-backward districts. The states Bihar and Uttar Pradesh each host over a quarter of the backward districts, while another quarter or so are located in Madhya Pradesh and Rajasthan. Fig. 1 shows their spatial distribution in Madhya Pradesh, one of India's largest states.

Backward districts are largely concentrated in the northern, north-eastern and central parts of India. Besides, backward districts also formed a belt across the states Rajasthan and Gujarat in the northwest. Notably, advanced states in the west or south such as Maharashtra and Karnataka also have a few backward districts, suggesting considerable spatial disparities existing within subregions of the country.

The program, as stipulated in Section 80-IA of Income Tax Act, offered new industrial enterprises in the backward districts a tax holiday in which 100 per cent of their profits would be exempt from income tax for the first five assessment years. This applied regardless of ownership structure. After the initial five assessment years, 30 per cent of profits would be exempt from tax in the case of companies (i.e., private or public enterprises registered under the Companies Act) for an additional five years, and 25 per cent in the case of unincorporated enterprises (such as sole proprietorships and partnerships). For enterprises owned by cooperative societies, 25% per cent of profits would be exempt from tax for a period of seven years after the initial five year period. To be eligible for the benefits, the industrial undertaking had to "begin to manufacture or produce articles or things or to operate its cold storage plant or plants at any time during the period beginning on the 1st day of October 1994 and ending on the 31st day of March 2000." The program excluded a few industries or economic activities from receiving tax exemption such as manufacture of products of tobacco, alcohol spirits, confectionery, and aerated waters.

Given that the corporate tax rate ranged from 45% to 40% over a good part of the 1990s, the tax holiday was substantial for public and private limited companies in the manufacturing sector.[3] Other types of manufacturing enterprises, such as proprietorships and partnerships were generally taxed at lower rates, especially those generating less income. In 1992–93, for instance, the tax rates were set at 20% for incomes between Rs30,000 to Rs50,000; 30% for incomes between Rs50,000 to Rs100,000; and 40% for incomes above Rs100,000.[4] In this way, the tax benefits of the backward district program can be expected to have been less attractive for smaller firms generating lower incomes for their owners.[5]

3. Data

We use establishment-level data from India's Economic Census of 1998 and 2005 (EC 1998 and EC 2005 thereafter), and district-level data from the Primary Census Abstract of 1991 (PCA 1991), in conjunction with the districts' gradation scores published in the "All India Gradation List" of the Income Tax Act, to evaluate the impacts of the backward districts program on local industrial development.[6] The EC, administered by the Central Statistical Organisation, Ministry of Statistics and Programme Implementation since 1977, provides a country-wide census of establishments engaged in all economic activities excluding crop production and plantations. The Primary Census Abstract (PCA) is a district-level data complied based on the Indian census data collected every decade.

EC 1998 contains key data on 27.9 million establishments from both rural and urban areas. For each establishment, we know the number of employees, major economic activity classified according to 1987 four-digit National Industry Classification (NIC 1987), location in terms of district and sub-district (e.g. towns and rural blocks), registration status, and so on.

We collapse the micro-data to obtain the total number of firms and employment at 2-digit industry by district level for our analysis. We classify 68 2-digit industries into 16 eligible industries, which include 9 heavy and 7 light manufacturing industries, and 52 ineligible industries from primary, mining, construction, utilities and services sectors. Heavy industries tend to be capital intensive and use heavy and bulky raw-materials in production. Examples are the manufacture of basic metals and alloys, metal products and parts, and machinery and equipment, including transport equipment. Light industries tend to be labor intensive and use relatively light raw-materials. The manufacture of food products, textiles, apparel, leather goods and footwear are prominent examples.

In addition, we create an "excluded" category that contains all 4-digit

[2] As per Section 3d of the Industries (Development and Regulation) Act of 1951, 'industrial undertaking' pertains to a scheduled industry carried on in one or more factories by any person or authority including the government.

[3] Corporate income tax accounted for 54% of total direct taxes in India in 1994–95.

[4] The various thresholds and tax rates were adjusted over time. For example, in 1997–98 the three tax brackets were reduced to 10%, 20%, and 30%.

[5] It should be noted that there is no one-to-one correspondence between formality and being part of the corporate sector. For example, an examination of Annual Survey of Industries data for 2004-05—a survey of enterprises registered under the Indian Factories Act and thus deemed to be "formal" in most discussions— reveals that 61% of firms were either proprietorships or partnerships as compared to 34% public and private limited companies. Most informal sector enterprises would be proprietorships (with other types of ownership forms, such as cooperative societies and partnerships comprising just a couple of percent of the total).

[6] We use the terms establishment and firm interchangeably in this study.

Table 1
State-wise distribution of backward and non-backward districts.

State Name	Full Sample			Core Sample		
	Backward	Non-Backward	Total	Backward	Non-backward	Total
Andhra Pradesh	2	21	23	2	–	2
Bihar	33	9	42	4	3	7
Gujarat	3	16	19	2	–	1
Haryana	–	16	16	–	–	–
Karnataka	1	19	20	1	3	4
Kerala	2	12	14	–	1	1
Madhya Pradesh	18	27	45	8	10	18
Maharashtra	2	28	30	1	7	8
Orissa	6	7	13	2	3	5
Punjab	–	12	12	–	–	–
Rajasthan	12	15	27	4	4	8
Tamil Nadu	–	21	21	–	2	2
Uttar Pradesh	35	28	63	13	5	18
West Bengal	9	6	15	1	1	2
Total	**123**	**237**	**360**	**38**	**39**	**77**

Notes: Backward districts were those with gradation score below 500 plus three districts with gradation scores exceeding 500, i.e. Idukki (618) and Wayanad (583) from Kerala and Jalapaiguri (728) from West Bengal, which were either a "no industry" district or an inaccessible hill area district, or did not have a "railhead". Core sample contains districts with gradation scores ranging from 351 to 650.

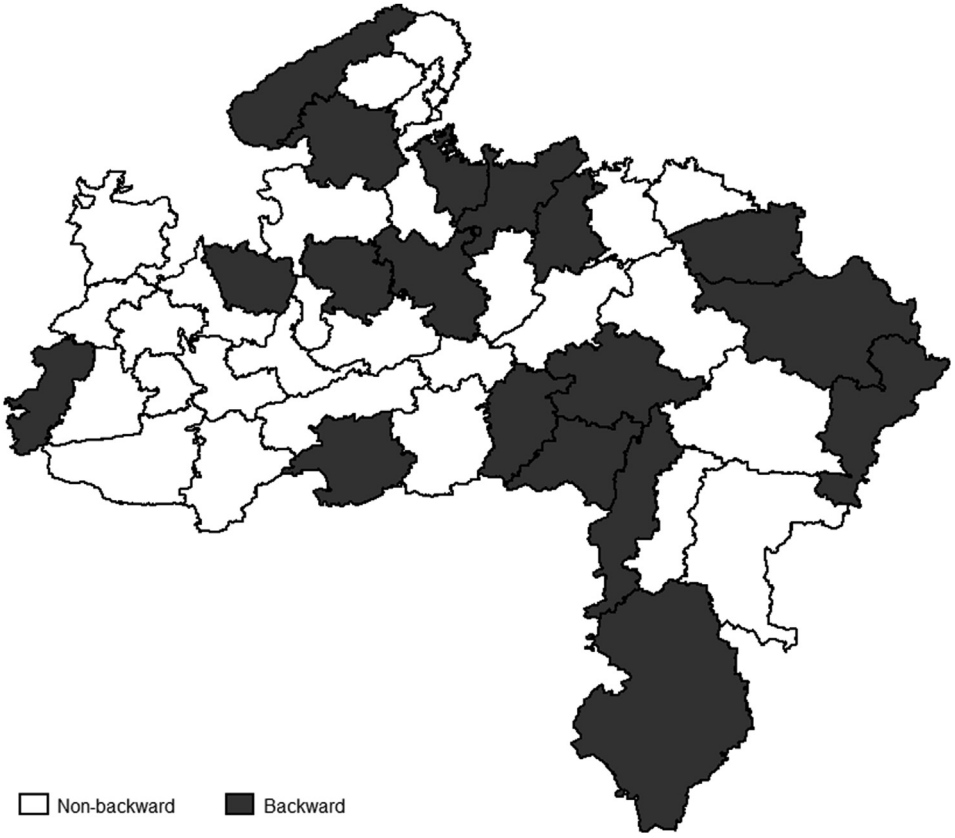


Fig. 1. Backward and non-backward districts in Madhya Pradesh (1991 Census Borders).

manufacturing industries that were either excluded outright or involved activities excluded from the program by the government but part of an eligible 2-digit industry.[7] Nevertheless, it is often the case that an excluded economic activity represents a subset of production within a 4-digit industry. For instance, the production of "latex foam sponge and polyurethane foam", which was listed as ineligible for the program, belongs to and accounts for a small portion of NIC 3020, or the "manufacture of plastics in primary forms; manufacture of synthetic rubber". Since we only observe data at the 4-digit NIC level, treating an industry like NIC 3020 as either eligible or ineligible would result in some measurement error. To address the issue, we adopt three definitions for the excluded category with increasing stringency in classifying a 4-digit NIC as an ineligible industry. The baseline analysis uses a "middle path" definition that contains 50 4-digit NICs. In robustness checks we examine both the more conservative definition (25 NICs) and more liberal definition (91 NICs).

[7] The full list of excluded items was specified in provisions of the 11th Schedule of the Income Tax Act (http://www.incometaxindia.gov.in/Acts/Income-tax%20Act,%201961/2013/102120000000027705.htm).

One potential issue with the use of EC 1998 is that it uses districts that can have different geographic boundaries from those in 1991, i.e., those used for defining backward districts by the program in 1994. With regards to the 14 states under consideration in our study, there are 100 more districts in EC 1998 as opposed to the 360 in 1991. Fortunately, the reorganization of districts does not nullify our identification strategy in that the backward status accorded in 1994 was carried forward to the newly appointed districts. We construct the data following the 1991 definition of districts so as to match them with the gradation scores. For the most common cases whereby an old district in 1991 was split into multiple ones by 1998, we simply collapsed the data of the new districts into the old district. For a few complex cases whereby a new district was formed by parts carved out from multiple old districts, we partition the data of the new district using population weights developed in Kumar and Somanathan (2009), and merge them back into their original districts.

Our final dataset was organized into 24,840 industry-by-district units formed by 360 districts and 69 2-digit industries. Twelve percent, or 3016 of the units are empty implying there were no firms and employment in those industry-by-district cells. In order to work with all the units in the analysis, we transform the number of firms and employment by log(Y+1).[8] As a robustness check, we also consider log(Y), thereby dropping district-industry units with zero observations from our analysis. Although a larger sample with more disaggregated units (e.g. district by 3-digit NIC) could be constructed from the EC, it comes at the expense of getting extremely high frequencies of zero observations. By keeping our analysis at 2-digit industry-by-district level, we strike a balance between having sufficient non-zero observations and adequate sample size.

Table 2 presents means and standard deviations of four variables of interest, namely numbers of firms, employment, share of formal firms and share of employment in formal firms, by light, heavy and other (ineligible) industries and district treatment status in 1998. Firms that registered under Factories Act 1948 or with other authorities such as various industry authorities (e.g. state departments of industry) are considered formal firms.[9]

As far as the full sample is concerned, the backward districts had on average substantially smaller number of firms, employment size, and share of formal businesses than the non-backward districts across different industries. For instance, the number of firms averaged across the light manufacturing industries and districts was 953 in the backward districts and 1208 in the non-backward districts; the average share of formal firms was 4.2% and these hired 8% workers in the heavy manufacturing industries in the backward districts, whereas the corresponding shares were 13% for firms and 27% for employment in the non-backward districts.

We further identify a core sample of districts with gradation scores falling between 351 and 650. There are almost equal numbers of districts – 38 backward districts and 39 non-backward – within the same bandwidth of 150 on each side of the cutoff.[10] Interestingly, when these subsets of backward and non-backward districts are compared, as shown in the right panel of Table 2, the backward districts had higher average number of firms and employment in the light and heavy manufacturing industries, and ineligible industries to a lesser extent, than the non-backward districts. On the other hand, the backward districts still had lower degree of formalization than the non-backward districts in the core sample across industry category, although the gaps were reduced considerably.

EC 2005 contains about 38.4 million establishments, 37.5% more than those in EC 1998. Similar steps were followed to process raw EC 2005 data. Districts in 2005 were mapped to 1991 districts. EC 2005 uses the NIC 2004 for industrial classification, which is more detailed in some sectors and compressed in others. We checked the descriptions of 2-digit and 4-digit NICs and matched them to NIC 1987 at 2-digit level as

Table 2
Summary statistics of firms and employment by industrial category and treatment status in 1998.

VARIABLES		Full Sample		Core Sample	
		Backward	Non-backward	Backward	Non-backward
Light Manufacturing					
Number of firms	*Mean*	953	1208	1179	963
	Std	3178	2648	3385	2105
Employment	*Mean*	2187	4614	2803	2528
	Std	7592	12,509	8021	6815
	Obs.	840	1659	266	273
Share of formal firms	*Mean*	0.025	0.078	0.032	0.033
	Std	0.071	0.149	0.088	0.088
Share of formal employment	*Mean*	0.047	0.179	0.062	0.071
	Std	0.119	0.25	0.141	0.154
	Obs.	810	1625	262	265
Heavy Manufacturing					
Number of firms	*Mean*	247	477	307	265
	Std	584	993	712	478
Employment	*Mean*	855	3032	1072	875
	Std	2324	7427	2865	1719
	Obs.	1080	2133	342	351
Share of formal firms	*Mean*	0.042	0.130	0.049	0.066
	Std	0.110	0.180	0.109	0.138
Share of formal employment	*Mean*	0.080	0.270	0.103	0.119
	Std	0.179	0.296	0.201	0.217
	Obs.	1023	2106	335	345
Ineligible Industries					
Number of firms	*Mean*	760	1512	980	960
	Std	2844	5359	3732	2992
Employment	*Mean*	1556	3782	2053	2044
	Std	5486	12,211	6824	5361
	Obs.	6360	12,561	2014	2067
Share of formal firms	*Mean*	0.019	0.045	0.017	0.036
	Std	0.072	0.124	0.067	0.123
Share of formal employment	*Mean*	0.028	0.066	0.028	0.048
	Std	0.098	0.155	0.093	0.141
	Obs.	5098	10,878	1669	1743

Note: Core sample contains 38 backward districts with gradation scores between 351 and 500 and 39 non-backward districts between 501 and 650. Light manufacturing contains seven 2-digit industries, heavy manufacturing nine industries, and ineligible industries 52 industries plus 1 excluded category. For reporting formalization rates, the industry-by-district cells with zero firms are dropped.

[8] As an alternative, we use an inverse hyperbolic sine transformation of the dependent variables. The results are literally the same (available upon request).

[9] Comparing firm registration information across EC 1998 and EC 2005 reveals that the registration rates of firms in 1998 were lower because EC 1998 missed a few categories such as "Shop and Establishment Act" and "other agencies" in the list of possible answers to the registration question. When the additional categories were added to EC 2005, a majority of registered firms were found under these. Nevertheless, the changes to the registration question is not that problematic. First, registration under the Factories Act—an option in both datasets—is probably the most relevant feature in determining formality in the Indian context. It applies to industry and is the channel through which India's labor regulations and industrial relations system (in addition to health and safety regulations) kick in and affect firms. Second, we mainly rely on cross-sectional comparisons to estimate the program's effects on formalization. To the extent that this systematic change to the definition of formal firms applies to the entire county, our estimates are still informative for both years considered separately.

[10] The necessity of focusing our analysis on the core sample is discussed in the succeeding section. The right panel of Table 1 presents distribution of the core sample districts across states.

precisely as we could. Seven two-digit light manufacturing industries and 9 heavy manufacturing industries are identified as eligible although they do not correspond one-to-one to those in 1998 data at 4-digit level. Non-manufacturing industries in NIC 2004 are typically more consolidated than NIC 1987, thus we only have 36 two-digit NICs plus one exclusion category belonging to the ineligible group. Descriptive statistics of EC 2005 are presented in Table 9 and discussed in Section 6.

4. Empirical strategy

The major identification challenge in evaluating the impact of the backward district program is that backward districts are likely to be distinct from the non-backward districts in both pre-program conditions and development trends without the program. However, the way the government identifies backward districts allows us to apply a regression discontinuity design to credibly estimate the causal effects of the program.

As described earlier, each of the 360 districts from 14 states considered under the program was assigned a composite score in 1993. With the exception of three districts, treatment status was determined strictly by whether the score is above the cutoff point of 500 or not. The formula to compute the score involves eight independent measurements as well as their national averages using various data from early 1990s. Thus, it is very hard to conceive of any scope for a state or district government to manipulate the score to make its districts or itself eligible for the program. To further verify this assumption, we inspect the density of the gradation score at the cutoff point in Fig. 2. There is no visible bunching of the graduation scores right below the cutoff value of 500, lending support to the claim that the districts had little chance to make themselves eligible for the program.

Another threat to our identification strategy is the possibility that other differences correlated with the outcomes of interest existed between the treated and untreated districts near the cutoff point. To check this, we examine whether there is any discontinuity present at the cutoff point for the pre-treatment district characteristics from PCA 1991 as suggested in Imbens and Lemieux (2008).

The left panel of Table 3 compares all backward districts against all non-backward districts. Not surprisingly, the differences between the two groups of districts are stark. Out of 11 variables considered, 6 are statistically significant. On average, the backward districts had smaller population, fewer main workers (who had worked 6 months or more in the survey year), fewer workers in manufacturing and in trade and commerce, fewer residential units, and lower literacy rate than the non-backward districts. The magnitude of these gaps are substantial. On the other hand, the backward districts are similar to non-backward districts in gender ratio, worker participation rate, and number of primary workers and marginal workers.[11]

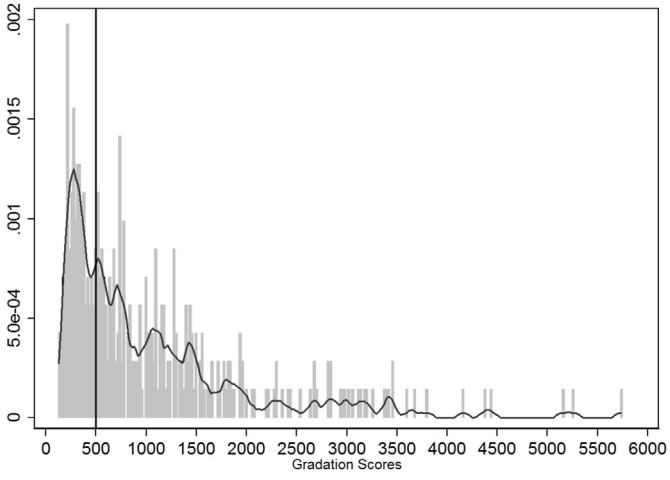


Fig. 2. Distribution of gradation scores. Notes: The bin size for bars is 40 points of gradation scores. The density curve is estimated with epanechnikov kernel and bandwidth of 40 points.

However, the right panel of Table 3 shows that when the band for comparison is reduced to include districts from the core sample only, none of the variables is statistically different between the treated and untreated districts. More importantly, the vanishing of statistical differences is mainly attributable to diminished differences in means (in absolute value) rather than the modest increases in standard errors due to smaller sample sizes.

Take the number of manufacturing workers as an example, which may be of particular interest as it is closely related to our outcome variables and measured before the policy intervention. The average number of manufacturing workers across all backward districts was 30,600, about 30% of the average over all non-backward districts in 1991. When we focus on the core sample, however, the two means get remarkably close with a 6% difference in values and p-value equal to 0.783. This implies that the two groups of districts in the core sample had similarly sized manufacturing sectors before the program.

Overall, the exercise confirms that there is no discontinuity detected in the mean values of a set of pre-treatment characteristics for districts with a neighborhood surrounding the threshold of 500. Thus, the variation in treatment status could be considered as good as randomized, and the RD approach should yield credible estimates of the program's impacts for the core sample (Lee and Lemieuxa, 2010). On the other hand, the RD estimation applied to a broader sample or full sample of districts may not be informative given substantial differences in initial conditions between the backward and non-backward districts in the expanded samples. In other words, the RD strategy may only be effective in providing consistent estimates of a local average treatment effect (LATE). In view of this, we focus our analysis on the core sample hereafter while providing a brief discussion of the baseline results for the expanded samples.[12]

We estimate three specifications of parametric RD regressions in the baseline analysis. The first is of the form:

$$Y_{id} = \beta_0 + \beta_1 T_d + f(Z_d) + X_d^{1991}\varphi + \eta_i + \eta_s + \varepsilon_{id} \quad (1)$$

where Y_{id} is log of number of firms or total employment (plus one) or share of formal firms or employment in formal firms in 2-digit industry i of district d. T_d is a binary indicator equal to 1 if district d is designated backward. Following Gelman and Imbens (2019), we use a quadratic polynomial function for $f(Z_d)$, where Z_d is the normalized gradation score equal to the raw score divided by 500 minus 1. X_d^{1991} denotes a vector of district-level variables obtained from the 1991 population census including area, population, numbers of main workers, primary workers and manufacturing workers, all expressed in logs, and the worker participation rate and literacy rate. η_i and η_s are 2-digit industry and state fixed effects, respectively. β_1 measures the aggregate effect of the program on the backward districts across all industries.

The second specification distinguishes eligible industries from ineligible industries:

$$Y_{id} = \beta_0 + \beta_2 T_d \times S_i + \gamma T_d \times (1 - S_i) + f(Z_d) + X_d^{1991}\varphi + \eta_i + \eta_s + \varepsilon_{id} \quad (2)$$

where S_i is a binary indicator equal to 1 if industry i is a manufacturing industry eligible for the program. While the program targeted

[11] We also narrow the comparison to a restricted sample, which consists of 82 backward districts and 80 non-backward districts with gradation scores between 251 and 850, excluding extremely underdeveloped and advanced districts. The mean differences between the backward and non-backward districts remain statistically and economically significant.

[12] The strategy to focus on core sample bears similarity to Greenstone et al. (2010), which restricted the treatment and control groups to winning counties and barely "losing" counties of the new Million Dollar Plant.

Table 3
T-tests of pre-treatment district variables from PCA 1991

Variable	Full Sample				Core Sample			
	T = 0 (N = 237)	T = 1 (N = 120)	Diff.	p-value	T = 0 (N = 39)	T = 1 (N = 38)	Diff	p-value
Population (in 1000)	2341.7	1901.3	−440.4	0.005	1882.3	2050.5	168.1	0.517
	(101.2)	(94.1)	(156)		(159.0)	(202.7)	(258.4)	
Main Workers (in 1000)	820.7	610.1	−210.6	<.001	641.7	678.8	37.1	0.649
	(34.3)	(28.5)	(51.8)		(49.4)	(64.2)	(81.3)	
Marginal Workers (in 1000)	73.1	73.5	0.4	0.954	76.4	78.1	1.7	0.902
	(3.8)	(5.2)	(6.5)		(8.2)	(11.0)	(13.8)	
Number of occupied residential houses (in 1000 units)	423.4	306.8	−116.6	<.001	321.1	333.1	12.0	0.783
	(19.6)	(15.4)	(29.4)		(26.5)	(34.2)	(43.4)	
Workers – Agri. Fishing Farming (in 1000)	733.0	662.9	−70.1	0.154	680.8	707.8	27.0	0.769
	(31.2)	(31.8)	(49.0)		(59.2)	(69.7)	(91.6)	
Workers - Manufacturing (in 1000)	99.1	30.6	−68.6	<.001	37.2	39.7	2.4	0.783
	(8.8)	(3.3)	(12.5)		(5.4)	(6.8)	(8.7)	
Workers - Trade and Commerce (in 1000)	69.4	26.4	−43.0	<.001	33.8	32.6	−1.1	0.845
	(5.8)	(1.9)	(8.2)		(4.0)	(4.2)	(5.8)	
Area (square kilometers)	13.6	8.3	−5.2	0.344	6.3	5.6	−0.8	0.528
	(3.5)	(3.6)	(5.5)		(0.7)	(0.9)	(1.2)	
Worker participation rate (%)	38.37	37.72	−0.65	0.406	39.36	38.12	−1.24	0.431
	(0.45)	(0.67)	(0.78)		(1.09)	(1.11)	(1.56)	
Literacy rate (%)	54.95	39.38	−15.56	<.001	47.07	45.23	−1.84	0.534
	(0.94)	(0.98)	(1.48)		(2.12)	(2.05)	(2.94)	
Number of females (per 1000 males)	93.30	92.52	−0.78	0.243	93.56	92.69	−0.87	0.534
	(0.40)	(0.51)	(0.66)		(1.05)	(0.91)	(1.39)	

Notes: T refers to treatment group; T = 1 are the backward districts and T = 0 are non-backward districts. Core sample contains districts with gradation scores ranging from 351 to 650. Three backward districts with gradation scores above 500 are excluded from the full sample comparison. Standard errors are in parenthesis.

manufacturing industries, other industries in the backward districts could have experienced positive or negative spillovers. The former may take place through input-output linkages or other agglomeration-related channels, while the latter through increased local competition for inputs such as land and credit. We expect β_2 to be positive if the program boosted manufacturing in backward districts, and γ to be signed either way or even zero.

Our third specification explores differential effects of the program on light and heavy manufacturing:

$$Y_{id} = \beta_0 + \beta_3 T_d \times S_i^l + \beta_4 T_d \times S_i^h + \gamma T_d \times (1 - S_i) + f(Z_d) + X_d^{1991}\varphi + \eta_i + \eta_s + \varepsilon_{id} \quad (3)$$

where S_i^l (S_i^h) is dummy equal to 1 if industry i is one of the eligible light (heavy) manufacturing industries as described in the data section. Relative to heavy manufacturing productions, light manufacturing requires smaller capital investment, fewer skilled workers and application of less sophisticated technology. Hence, light manufacturing industries often get developed ahead of heavy manufacturing in underdeveloped regions. It seems reasonable to conjecture that the program was more effective in promoting growth in light manufacturing than in heavy manufacturing in backward districts. In this case, we expect that $\beta_3 > \beta_4 \geq 0$.

5. Program impacts in short term

5.1. Economic activity

We first examine how the program affected local economic activity as measured by the number of firms and employment. Before reporting the RD estimation results, we show plots of the log transformed data by industry category in Fig. 3. Each dot represents mean counts of firms or employment averaged across 2-digit industries within each category (i.e. light manufacturing, heavy manufacturing and ineligible industries) and districts falling in a bin of size of 40 gradation scores. The solid line represents the local polynomial fit of the data with degree 1 and bandwidth equal to 200. The dashed lines represent the 95% confidence interval based on the local polynomial estimation.

In all six plots, we see that the solid lines rise with the gradation score on both sides of the cutoff point. Meanwhile, there is a downward break between the two lines around where the gradation score crosses the cutoff. The gaps are considerably larger for light manufacturing (top panel) than for heavy manufacturing (middle) and ineligible industries (bottom). Once we scrutinize the graphs, we can see the breaks are largely due to a warping up of the segment of the left-hand lines near the cutoff point. Finally, the patterns of firm counts and employment resemble each other. Overall, Fig. 3 suggests that the program had a positive impact on firm entry and employment in backward districts, especially those closer to the cutoff point.

Table 4 reports the results from estimating equations (1)–(3) for the core sample with log-transformed firm counts and employment as dependent variables. Columns (1) and (4), corresponding to model in equation (1), show that the point estimates are 0.114 for number of firms and 0.133 for employment. This implies that the program may have had some positive aggregate effect on the economy of backward districts near the cutoff. However, this is far from conclusive since the estimates lack statistical precision.

When the eligible and ineligible industries are separately considered as shown in columns (2) and (5), both the number of firms and employment in eligible manufacturing industries in the backward districts are 36% higher than those in non-backward districts. The coefficient estimates are statistically significant at 1% or 5% level. The estimates suggest that the manufacturing sector had grown 36%, or 8% per annum, faster in the treated districts than the untreated districts over four years of program implementation. Suppose the average increase of manufacturing was 5% per annum in non-backward districts of the core sample. The average increase in backward districts would be 13.4% (i.e., (1 + 5%)*(1 + 8%)-1). Lastly, the estimates for ineligible industries are small and statistically insignificant, suggesting that the program's agglomeration spillovers to the ineligible industries were limited.

Columns (3) and (6) show evidence of heterogeneous effects of the program on light and heavy manufacturing productions. Compared to non-backward districts near the cutoff, the light manufacturing in

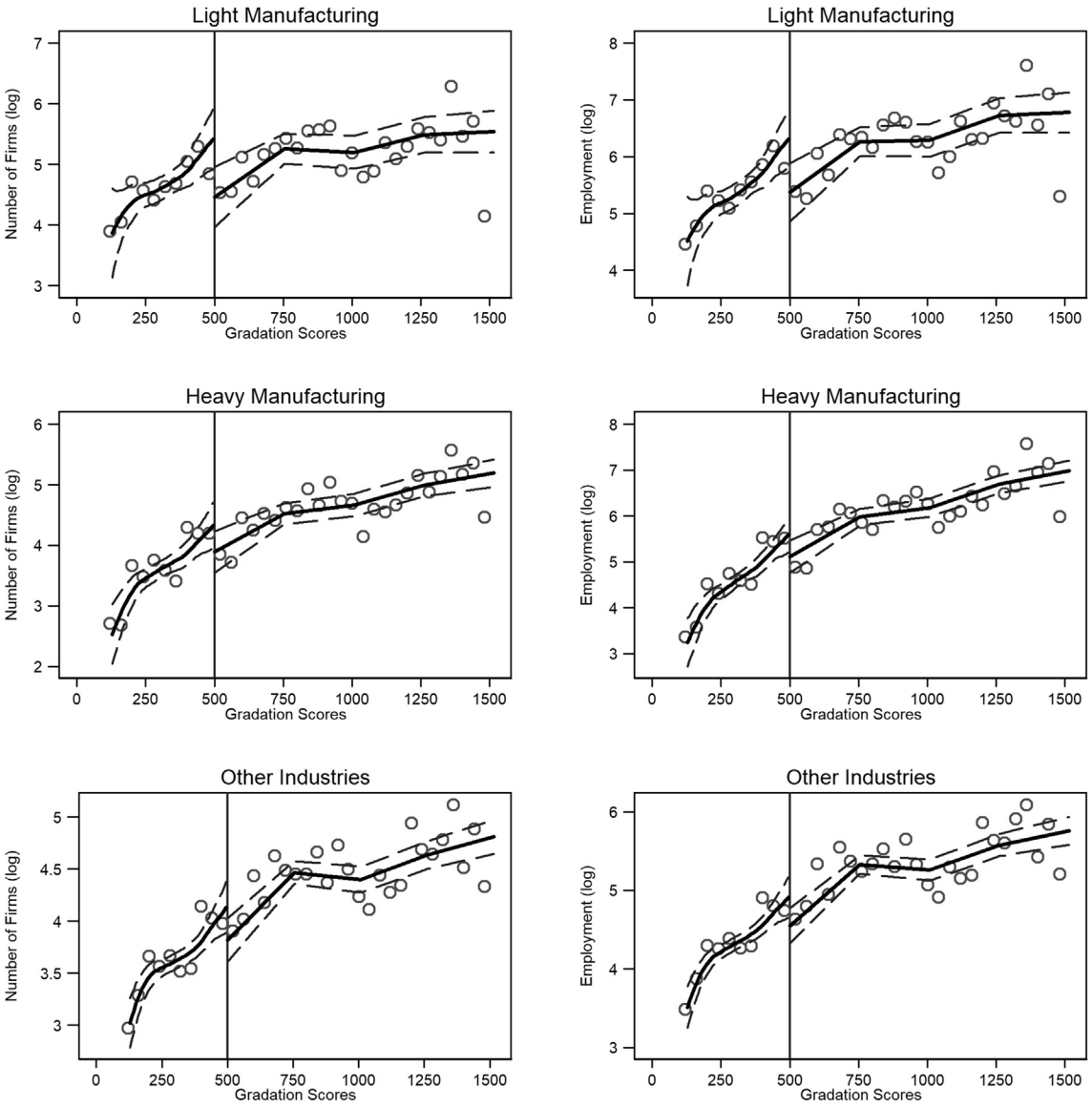


Fig. 3. Number of firms and employment of 2-digit industry by district relative to the gradation scores in 1998. Notes: Each dot represents mean counts of firms or employment averaged across 2-digit industries within the category (i.e. light manufacturing, heavy manufacturing and ineligible industries) and districts falling in a bin of size of 40 gradation scores. The solid line is a local polynomial fit with degree 1 and bandwidth equal to 200. The dashed lines represent the 95% confidence interval of the local polynomial.

backward districts grew 59% more in both number of firms and employment, which translates into an additional 12.3% increase per year over the four year period. On the other hand, the effects on heavy manufacturing are estimated around 18% without statistical significance. This means that the effects of the program on manufacturing in the backward districts are mostly driven by the effects on light manufacturing industries. The results support our conjecture that the tax incentives offered by the program are more likely to induce development in light manufacturing while it is harder to promote heavy manufacturing sectors, which face more complex and severe constraints in underdeveloped areas.

It is also interesting to note that the estimated effects are almost identical for number of firms and employment, suggesting little change in the average firm size in the better-off backward districts. This implies that the program's employment impacts are driven by firm entry rather than growth of existing firms, which is consistent with program targeting its incentives at new firms.

As far as the control variables are concerned, the patterns seem sensible and consistent across the dependent variables. The polynomial functions show weak concave or linear shapes within the core sample score range. Pre-treatment district-level covariates other than the number of main workers are positively correlated with the dependent variables, with literacy rate, numbers of manufacturing workers and primary workers, and district area statistically significant for one or both dependent variables. Controlling for these pre-program district characteristics improves the credibility of our estimation on program effects.

In sum, Table 4 shows that the program had generated considerable positive impacts on local manufacturing sector in the backward districts which had the highest gradation scores or best initial conditions for development among all backward districts. Moreover, as expected, light

Table 4
Program impacts on number of firms and employment in 1998.

VARIABLES	Number of firms			Employment		
	(1)	(2)	(3)	(4)	(5)	(6)
Backward district	0.114			0.133		
	(0.087)			(0.099)		
Backward district * eligible manufacturing		0.360***			0.360**	
		(0.111)			(0.138)	
Backward district * eligible light manufacturing			0.592***			0.590***
			(0.167)			(0.199)
Backward districts * eligible heavy manufacturing			0.180			0.181
			(0.114)			(0.140)
Backward district * ineligible industries		0.040	0.040		0.064	0.064
		(0.094)	(0.094)		(0.105)	(0.105)
Function of Gradation Score						
Linear score	0.480	0.480	0.480	0.701*	0.701*	0.701*
	(0.301)	(0.301)	(0.301)	(0.353)	(0.353)	(0.354)
Squared score	−0.327	−0.327	−0.327	0.092	0.092	0.0922
	(0.903)	(0.903)	(0.903)	(1.047)	(1.047)	(1.047)
1991 District Characteristics:						
Area (log)	0.127	0.127	0.127	0.172*	0.172*	0.172*
	(0.084)	(0.084)	(0.084)	(0.098)	(0.098)	(0.098)
Population (log)	0.723	0.723	0.723	0.610	0.610	0.610
	(0.445)	(0.445)	(0.445)	(0.551)	(0.551)	(0.551)
Worker participation rate	0.015	0.015	0.015	0.010	0.010	0.010
	(0.010)	(0.010)	(0.010)	(0.012)	(0.012)	(0.012)
Literacy rate	0.010***	0.010***	0.010***	0.011***	0.011***	0.011***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Primary workers (log)	0.667**	0.667**	0.667**	0.442	0.442	0.442
	(0.315)	(0.315)	(0.315)	(0.311)	(0.311)	(0.311)
Manufacturing workers (log)	0.243***	0.243***	0.243***	0.180***	0.180***	0.180***
	(0.045)	(0.045)	(0.045)	(0.035)	(0.035)	(0.035)
Main workers (log)	−0.903	−0.903	−0.903	−0.400	−0.400	−0.400
	(0.669)	(0.669)	(0.669)	(0.722)	(0.722)	(0.722)
State dummy	Y	Y	Y	Y	Y	Y
2-digit industry dummy	Y	Y	Y	Y	Y	Y
Observations	5313	5313	5313	5313	5313	5313
R-squared	0.827	0.827	0.828	0.794	0.795	0.795

Notes: The core sample including backward districts with gradation scores ranging from 351 to 650 is used in the regressions. The dependent variable was transformed as log(Y+1). Gradation score is normalized by dividing the original score with 500 and then subtracting 1. Standard errors in parentheses are clustered at district level. *** indicates significance at 1%, ** significance at 5%, * significance at 10%.

manufacturing industries had benefited from the program much more greatly than had heavy industries in the same areas.[13]

5.2. Robustness tests

We conducted a variety of robustness tests to check sensitivity of our baseline results to the variations in sample treatment and estimation approach. Table 5 reports the main test results with equation (3) estimated for the core sample.

First, we apply a simple log transformation to the number of firms and employment, i.e. log(Y), instead of the log(Y+1) transformation used in the baseline. This leads to a 12% reduction in the effective sample. However, the coefficient estimates for light manufacturing, heavy manufacturing and ineligible industries only change slightly (Column 1). Thus, excluding the zero-valued units in the analysis does not alter our main finding that the program had a large positive impact on light manufacturing in the better-off backward districts.

Second, as explained in the data section we construct an excluded category of 4-digit industries that contains some ineligible manufacturing activity. The scope of this category depends on how aggressively we consider such a 4-digit industry as ineligible for the program as a whole. Note that a change made to the excluded category will cause a corresponding change to the scope of eligible industries, and consequently changes to the dependent variables of many sample units. Using two alternative definitions of the excluded category, one more conservative and the other more liberal than the one used for the baseline, we obtain virtually identical estimates (Columns 2 and 3).

Third, about 22 million or 79% of firms in EC 1998 are small firms with 1 or 2 employees. To examine whether our results are driven by these small, mostly self-employed, firms, we re-construct our sample at the 2-digit industry by district level with only firms with 3 or more employees. The estimated program effects on light manufacturing and heavy manufacturing industries increase slightly for both firm counts and employment (Column 4). Interestingly, the estimates on the ineligible industries increase by 2 or 3 times as compared to the baseline estimates with statistical significance at 10% level for firm counts, respectively. This implies that positive spillovers across industries may exist amongst relatively big firms.

With respect to other treatments of the sample, we estimate models with the four states, Bihar, Madhya Pradesh, Rajasthan and Uttar Pradesh, dropped one at a time. Since these four states each account for a large share of the sample, this helps us check whether the results are driven by any single state. Second, we drop the districts which neighbor the backward states and union territories receiving similar policy support under the Finance Act of 1993, which may have generated some spillovers over the state borders. Third, we exclude districts with boundary

[13] We also estimate equations (1)–(3) for the full sample as well as a restricted sample with gradation scores between 251 and 850. The coefficient estimates of interest diminish and turn negative in some cases (see Table A2). This could arise if the program mostly benefited the better-off backward districts with the disadvantages of the more underdeveloped districts too deep to be offset by the tax relief only. However, in light of the significant gaps in initial conditions between backward and non-backward districts far away from the cutoff, the estimates with the expanded samples are unlikely to be informative. Hence, we focus on the core sample throughout the paper.

Table 5
Robustness checks.

VARIABLES	Log(Y)	Conservative exclusion	Liberal exclusion	No small firms	Polynomial int. with backward dummy	3rd Order Polynomial	S.E. state clustering	Incl. other policies	False cut-off at 350
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Number of firms									
Backward district * eligible light manufacturing	0.537***	0.601***	0.593***	0.641***	0.624***	0.678***	0.592***	0.582***	−0.0887
	(0.166)	(0.168)	(0.167)	(0.167)	(0.179)	(0.184)	(0.164)	(0.164)	(0.163)
Backward districts * eligible heavy manufacturing	0.165	0.190*	0.177	0.241**	0.213*	0.267**	0.180	0.171	0.0601
	(0.112)	(0.113)	(0.114)	(0.116)	(0.118)	(0.125)	(0.106)	(0.117)	(0.165)
Backward district * ineligible industries	0.0137	0.0374	0.0411	0.170*	0.0726	0.126	0.0400	0.0301	0.117
	(0.0952)	(0.0943)	(0.0931)	(0.0876)	(0.0904)	(0.101)	(0.0874)	(0.0826)	(0.115)
Observations	4619	5313	5313	5313	5313	5313	5313	5313	5658
R-squared	0.783	0.825	0.828	0.795	0.828	0.828	0.828	0.828	0.828
Employment									
Backward district * eligible light manufacturing	0.508***	0.593***	0.597***	0.646***	0.612***	0.661***	0.590**	0.579**	−0.124
	(0.181)	(0.200)	(0.199)	(0.223)	(0.206)	(0.208)	(0.197)	(0.195)	(0.184)
Backward districts * eligible heavy manufacturing	0.158	0.185	0.173	0.233	0.203	0.252*	0.181	0.170	−0.0487
	(0.136)	(0.137)	(0.138)	(0.159)	(0.144)	(0.143)	(0.146)	(0.151)	(0.176)
Backward district * ineligible industries	0.0368	0.0607	0.0699	0.168	0.0865	0.136	0.0644	0.0533	0.163
	(0.0988)	(0.106)	(0.104)	(0.110)	(0.0974)	(0.105)	(0.119)	(0.114)	(0.121)
Observations	4619	5313	5313	5313	5313	5313	5313	5313	5658
R-squared	0.729	0.794	0.796	0.753	0.795	0.795	0.795	0.795	0.798
State dummy	Y	Y	Y	Y	Y	Y	Y	Y	Y
2-digit industry dummy	Y	Y	Y	Y	Y	Y	Y	Y	Y
2nd order polynomials of gradation score	Y	Y	Y	Y	Y	Y	N	Y	Y
District covariates	Y	Y	Y	Y	Y	Y	Y	Y	Y

Notes: The core sample including backward districts with gradation scores ranging from 351 to 650 is used throughout except in column (9), which uses a sample of districts with gradation scores between 251 and 500. The dependent variable was transformed as log(Y+1) except in column (1). Gradation score is normalized by dividing the original score with 500 and then subtracting 1. Standard errors in parentheses are clustered at district level except in column (5). *** indicates significance at 1%, ** significance at 5%, * significance at 10%.

changes between 1991 and 1998, which leaves us with 61 districts in the core sample (30 backward and 31 non-backward). Again, these exercises do not result in changes to our baseline estimates.[14]

In terms of alternative estimation approaches, we first try non-parametric estimation recently developed in Calonico et al. (2014) and Calonico et al. (2020). The treatment effects estimated for light manufacturing, heavy manufacturing and ineligible industries, respectively, are 1.032 (0.368), 0.442 (0.283), and 0.314 (0.200) for number of firms and 1.010 (0.380), 0.614 (0.317), and 0.391 (0.214) for employment. They are in line with, though greater than, the estimates from the parametric estimates.[15]

We allow the quadratic polynomial function to have different coefficients for backward and non-backward districts in column (5) and use cubic polynomial function in column (6). The estimates for light and heavy manufacturing increase moderately and those for heavy manufacturing turn statistically significant at 10% or 5% level.

We estimate models with standard errors clustered at the state level, which is more conservative for statistical inference if the district-level unobservables were positively correlated across districts within a state. The effects on light manufacturing remains significant at 5% or 1% level (Column 7).

Furthermore, we consider other place-based policies that the national and state governments of Indiaimplemented to promote the economy during the 1990s. Had some of them also been targeted on the backward districts, the treatment status in our sample could be contaminated. To address this concern, we identified three large-scale policies that implemented similar industry-specific interventions and overlapped in part with our district sample.

The first such policy is the Transport Subsidies Scheme, which provided transport subsidies from 50% to 90% up to five years to industrial units for the cost incurred on the movement of raw material and finished goods from the rail/ports to the location of the industrial unit. The scheme covered the hilly districts of Uttarakhand, Darjeeling of West Bengal and 13 other states and union territories, of which 10 districts belonging to 2 states were included in the backward district program.

The two other policies we considered, the Development of Growth Centers and the Integrated Infrastructure Development for Small Scale Industries, were concentrated on certain areas within a state or district. The Development of Growth Centers covered 71 growth centers across 30 states, of which 53 centers were located in our sample districts. These areas received Rs 250–300 million from the central government from 1992 to 1997 for developing the missing infrastructure necessary to attract industries. The Integrated Infrastructure Development for Small Scale Industries covered 121 centers from 26 states, with the centers selected from districts where infrastructure was a leading constraint to industrial development. The central government allocated Rs 50 million per center for infrastructural facilities related to power, water, roads, drainage, and banking from 1994 to 2009. Seventy-eight districts in our sample host such centers.

We create three dummies to indicate whether a district was treated by these policies, respectively, and control for them in the regression models. Column (8) shows that we obtain virtually the same estimates of the program effects. This is not unexpected since the treatment

[14] The results of these robustness checks are presented in Supplementary Appendix 1.

[15] We first use Stata package "rdbwselect" to obtain the optimal bandwidth. Subsequently, we apply "rdrobust" (2017 version) to obtain local point estimates while letting the algorithm select the local polynomial and order-bias of the non-parametric regression (Calonico et al., 2017). The optimal bandwidth varies between 121 and 190 depending on the industry category and dependent variable, which is largely consistent with our core sample with bandwidth of 150 on each side of the cutoff. Detailed estimates are provided in Supplementary Appendix 2.

assignment mechanism of the backward district program is unique and the treatment status is unlikely to be correlated with those of other policies under consideration. Moreover, the influence of the Development of Growth Centers and the Integrated Infrastructure Development for Small Scale Industries could be quite local without much scope for district-wide spillovers.

Finally, we undertake a falsification test to see whether a similar discontinuity in the outcome variables exists at a hypothetical cutoff. To mimic the setting of our primary estimates, we choose a gradation score equal to 350 with 44 backward districts to its left and 38 backward district to the right. The results in Columns 9 reveal that there is no jump in firm counts or employment at this hypothetical cutoff, lending further support to the program impacts we find at the real cutoff point.

5.3. Program impacts on formalization

The preceding subsections suggest that the backward districts program contributed to growth of manufacturing in the better-off treated districts, with the estimated effects on both the number of firms as well as employment on eligible industries remarkably robust. One question that arises is whether these findings might reflect the possibility that the program (inadvertently) led existing enterprises in backward districts to be better captured by the economic census. While the census is meant to capture all enterprises in covered sectors, in practice, capturing informal sector firms—i.e., typically small firms that are not registered with key statutory agencies—comprehensively in surveys is challenging. If tax avoidance is a key consideration that leads firms to avoid registration, it is possible that the program may have incentivized some existing informal firms in backward districts to register, and appear in the economic census by virtue of their having registered. While most observers believe that the decision to remain informal is driven more by the desire to avoid coming under the purview of the Factories Act—and thus avoid coming under the purview of India's regulatory apparatus governing labor, industrial relations, and worker health and safety issues—it is important to examine this possibility.

We estimate equations (1)–(3) with shares of formal firms and shares of employment in formal firms as dependent variables.[16] Table 6 summarize the results. In general, the program had a positive impact on the formalization rates in the backward districts across different industries with the point estimates slightly larger for light manufacturing. However, none of the estimates are statistically significant.[17] Taken literally, the point estimates suggest that most of the estimated program impacts are associated with real gains in manufacturing in backward districts, rather than from more pre-existing firms getting included in EC 1998. Meanwhile, the lack of statistical precision in the estimates prevents us from concluding that the program had improved formalization rates in the treated districts. This may echo the literature that finds informality to be quite persistent in some economies and that most policies with the objective of formalizing informal firms achieve only a modest increase in formalization (Bruhn and McKenzie, 2014; Piza, 2018).

5.4. Spatial spillovers

A place-based policy could cause positive spatial spillovers if the policy-induced production increases the demand for inputs and services across borders. On the other hand, capital and entrepreneurs are footloose. To access the benefits of the policy directly, they may choose to set up firms in or move existing firms to policy-treated areas. Such displacement is likely to occur in areas near treated areas, where other conditions are similar and the costs of border-crossing are not high. While we have identified significant positive effects of the backward district program on local manufacturing development, its actual contribution to nationwide economic growth would be larger or smaller depending on whether the program generated net positive or negative spillovers.

We investigate such spatial spillover effects in two ways. First, we split the core sample into two subsamples. One subsample consists of backward and non-backward districts sharing borders with at least one district from the other group; we refer to this as the "neighboring" subsample. The second consists of backward and non-backward districts that do not have common borders with any district from the other group and is referred to as the "non-neighboring" subsample. We end up with 24 backward and 24 non-backward districts in the neighboring subsample, and 14 backward and 15 non-backward districts in the non-neighboring subsample. Equation (3) is then estimated using each subsample. Should there exist net positive spillovers from the treated districts to the untreated, the program effects would be greater with spillover effects suppressed. Thus, the estimates with the non-neighboring subsample would be greater than those obtained with the whole core sample or neighboring subsample. The reverse holds true if net spatial spillovers are negative.

Table 7 shows estimation results from this exercise. For the neighboring subsample, the number of firms and employment in backward districts increased by 74% and 84%, respectively, for light manufacturing with statistical significance at the 1% level as compared to non-backward districts. Heavy manufacturing also grew by around 30% more, and ineligible industries by about 20%, in the backward districts with statistical significance at 5% or 10% level. These effects are greater than those estimated for the core sample as a whole. In contrast, the effects for the non-neighboring subsample are smaller with a 49% increase for number of firms and 37% for employment for light manufacturing industries.[18] Although the differences between the two subsample estimates are not statistically distinguishable, the gaps in the point estimates suggest that the program had net negative spillovers to the non-backward districts which are adjacent to a better-off backward district.

Next, we examine the spatial spillovers with a different approach, which compares districts with the same policy treatment status but differing in terms of their neighbor districts. Specifically, we compare backward districts that share borders with one or more non-backward districts against backward districts that share no borders with any non-backward district in the core sample. A similar comparison was also carried out among non-backward districts. In the presence of net negative spillovers, backward districts with non-backward neighbors would tend to benefit more than their peers without non-backward neighbors. Meanwhile, the non-backward districts with backward neighbors would be found worse off than the other non-backward districts. If positive spillovers dominated, however, the non-backward districts with backward neighbors could be better off than those without backward neighbors.

There may be concern that the backward (non-backward) districts adjacent to a non-backward (backward) district differ in some unobserved ways from those without a non-backward (backward) district in their neighborhood. We address this issue with two measures. First, we restrict the comparisons for backward and non-backward districts from the core sample only. That means that each sample falls in a narrow band of 150 graduation score points. Given the way it was developed, the gradation score is a good proxy for a district's conditions for business

[16] As noted earlier, firms that registered under Factories Act 1948 or with other authorities such as various industry authorities (such as state departments of industry) are considered formal firms.

[17] In view of the fact that a large share of industry-by-district cells had zero formalization rate, we also estimate equations (1)–(3) with a dummy dependent variable equal to 1 if the formalization rate is positive. The point estimates suggest that the program increased the probability of having at least one formal firm by 6–8 percent across industries in backward districts. However, none of them are statistically significant.

[18] The estimates are not statistically significant, which is probably due to reduced sample size causing inflated standard errors.

Table 6
Program impacts on formalization in 1998.

VARIABLES	Share of formal firms			Share of employment in formal firms		
	(1)	(2)	(3)	(4)	(5)	(6)
Backward district	0.036			0.045		
	(0.026)			(0.029)		
Backward district * eligible manufacturing		0.041			0.050	
		(0.030)			(0.035)	
Backward district * eligible light manufacturing			0.051			0.055
			(0.032)			(0.036)
Backward districts * eligible heavy manufacturing			0.033			0.046
			(0.029)			(0.038)
Backward district * ineligible industries		0.034	0.034		0.043	0.043
		(0.025)	(0.025)		(0.028)	(0.028)
State dummy	Y	Y	Y	Y	Y	Y
2-digit industry dummy	Y	Y	Y	Y	Y	Y
2nd order polynomials of gradation score	Y	Y	Y	Y	Y	Y
1991 district covariates	Y	Y	Y	Y	Y	Y
Observations	4619	4619	4619	4619	4619	4619
R-squared	0.232	0.232	0.232	0.203	0.203	0.203

Notes: The core sample including backward districts with gradation scores ranging from 351 to 650 is used in the regressions. The industry-by-district cells with zero firms are excluded. Gradation score is normalized by dividing the original score with 500 and then subtracting 1. Standard errors in parentheses are clustered at district level. *** indicates significance at 1%, ** significance at 5%, * significance at 10%.

Table 7
Program effects for backward districts (not) neighboring with non-backward districts.

VARIABLES	Neighboring subsample		Non-neighboring subsample	
	Firms	Employment	Firms	Employment
'Backward district * eligible light manufacturing	0.744***	0.842***	0.487	0.366
	(0.177)	(0.235)	(0.342)	(0.383)
Backward districts * eligible heavy manufacturing	0.282**	0.302*	0.164	0.173
	(0.118)	(0.168)	(0.286)	(0.330)
Backward district * ineligible industries	0.159*	0.214**	−0.004	0.014
	(0.084)	(0.094)	(0.192)	(0.231)
State dummy	Y	Y	Y	Y
2-digit industry dummy	Y	Y	Y	Y
2nd order polynomials of gradation score	Y	Y	Y	Y
1991 district covariates	Y	Y	Y	Y
Observations	3312	3312	2001	2001
R-squared	0.840	0.799	0.821	0.801

Notes: All districts are from core sample with gradation scores between 351 and 650. Neighboring sample consists of 24 backward and 24 non-backward districts, each with at least one neighboring district from the other group. Non-neighboring sample consists of 14 backward and 15 non-backward districts, none of which is neighboring a district from the other group. Gradation score is normalized by dividing the original score with 500 and then subtracting 1. Standard errors in parentheses are clustered at district level. *** indicates significance at 1%, ** significance at 5%, * significance at 10%.

development. We have also shown that backward and non-backward districts in the core sample were statistically identical in a number of pre-treatment characteristics. Therefore, differences between subgroups of the backward or non-backward districts from the core sample will be negligible.[19] Moreover, we control for the total number of backward districts and total number of non-backward districts (from the full sample) each district shares borders with in the regressions. The additional controls may help to capture unobserved geographical features that are correlated with industrial development at the district level, leaving the conditional differences in outcomes attributable to spatial spillovers.

We estimate the following regression for backward and non-backward districts from the core sample separately:

$$Y_{id} = \theta_0 + \theta_1 N_d^{opp} + \theta_2 W_d^T + \theta_3 W_d^U + f(Z_d) + X_d^{1991}\varphi + \eta_i + \eta_s + \varepsilon_{id} \quad (4)$$

where N_d^{opp} denotes whether district d has a district from the core sample with the opposite treatment status in its neighborhood. We also estimate equation (4) in which N_d^{opp} equals the number of districts from the core sample with opposite treatment status in the neighborhood of district d. W_d^T and W_d^U measure the total numbers of backward districts and non-backward districts in the neighborhood of district d, respectively.

Table 8 reports the estimates of θ_1 with number of firms and employment as dependent variables. On average, backward districts sharing borders with any non-backward districts in the core sample had 19% more firms and 10% more employment than backward districts without any non-backward district in the neighborhood. In contrast, non-backward districts sharing borders with any backward districts in the core sample had 26% fewer firms and 30% fewer employment than the other subgroup of backward districts. All estimates are statistically significant except that for employment in backward districts.

The pattern remains the same when the variable of interest is replaced with number of neighboring districts with the opposite treatment status. For instance, the number of firms would be 8% higher for a backward district with one more neighboring district from the non-backward group of the core sample than an otherwise similar backward district. Likewise, a non-backward district would lose 13% employment should the number of backward districts in its neighborhood increase by one. Again, the evidence from this second exercise indicates that negative spatial spillovers dominate the interaction between the treated and untreated districts.[20]

The two pieces of evidence we provide consistently points to net negative spillover effects associated with the backward district program. This is plausible as the backward and non-backward districts we focus on were highly similar in demographics and initial conditions for business

[19] The average gradation scores for 24 backward districts with non-backward neighbors and 14 without are 424.5 and 402.6, respectively. The average gradation scores for 24 non-backward districts with backward neighbors and 15 without are 560.2 and 565.8, respectively. These differences across subgroups are much smaller than those between the backward (416.4) and non-backward (562.4) districts of the core sample.

[20] We also estimate models with pooled core sample of districts and interactions between the backward district indicator, neighboring (non)-backward districts indicator (or counts) and industry type to evaluate spatial spillovers. The coefficients of interest all have the expected signs though lack statistical significance. Please see Supplementary Appendix 3 for more details.

Table 8
Effects of neighboring districts with opposite treatment status.

VARIABLES	Backward districts		Non-backward districts	
	Firms	Employment	Firms	Employment
Sharing borders with any district with opposite treatment status	0.192***	0.095	−0.259***	−0.300***
	(0.056)	(0.072)	(0.082)	(0.077)
Observations	2622	2622	2691	2691
R-squared	0.841	0.816	0.824	0.786
Number of districts with opposite treatment status sharing borders	0.081**	0.050	−0.082	−0.133**
	(0.033)	(0.044)	(0.062)	(0.056)
Observations	2622	2622	2691	2691
R-squared	0.841	0.816	0.823	0.785
Total numbers of adjacent backward districts	Y	Y	Y	Y
Total numbers of adjacent non-backward districts	Y	Y	Y	Y
State dummy	Y	Y	Y	Y
2-digit industry dummy	Y	Y	Y	Y
2nd order polynomials of gradation score	Y	Y	Y	Y
1991 district covariates	Y	Y	Y	Y

Notes: All districts are from core sample with gradation scores between 351 and 650. Twenty-four out of 38 backward districts share borders with at least one non-backward districts from core sample; 24 out of 39 non-backward districts share borders with at least one backward district from core sample. Standard errors in parentheses are clustered at district level. *** indicates significance at 1%, ** significance at 5%, * significance at 10%.

development. When they are also geographically close to each other, which implies limited costs of crossing district borders to set up a firm, the tax incentives provided in the program become attractive.[21] In this way, spatial displacement could partly offset the benefits of the program on the local economy when viewed from a regional or nationwide perspective.[22]

6. Long-term impacts of the program

While the backward districts program lasted only 10 years, it is important to see whether the program generated persistent effects on treated areas after it was phased out. As has been recognized in the literature (for example, Krugman, 1991), temporary policy shocks can have persistent effects. For example, an initial spurt in economic activities due to the program might generate self-sustaining agglomeration externalities.

We examine the long-term impacts of the program using Economic Census 2005 data. The program ended in 2004 for most firms other than those owned by cooperative societies (a fraction of firms, as noted earlier). Further, given that the program offered its most generous tax exemptions for a period of five years, the 2005 data should allow us to examine program effects after a significant reduction of program benefits—as much as 6 years for firms that availed of the program right at the outset—and one year after the program concluded. If the program had persistent effects, these should certainly show up in EC 2005 data.

Table 9
Summary statistics of firms and employment by industrial category and treatment status in 2005.

VARIABLES		Full Sample		Core Sample	
		Backward	Non-backward	Backward	Non-backward
Light Manufacturing					
Number of firms	*Mean*	1415	2438	1697	1663
	Std	3073	4493	3193	2853
Employment	*Mean*	2852	6836	3485	3243
	Std	7059	17,259	7621	6396
	N	840	1659	266	273
Share of formal firms	*Mean*	0.250	0.239	0.192	0.170
	Std	0.279	0.179	0.215	0.147
Share of formal employment	*Mean*	0.293	0.347	0.250	0.232
	Std	0.284	0.236	0.234	0.197
	N	785	1622	254	259
Heavy Manufacturing					
Number of firms	*Mean*	249	512	293	298
	Std	514	1298	621	505
Employment	*Mean*	774	2718	930	896
	Std	1852	6021	2146	1676
	N	1080	2133	342	351
Share of formal firms	*Mean*	0.321	0.380	0.279	0.270
	Std	0.291	0.234	0.247	0.214
Share of formal employment	*Mean*	0.404	0.542	0.389	0.378
	Std	0.306	0.281	0.287	0.267
	N	972	2032	313	325
Ineligible Industries					
Number of firms	*Mean*	1380	2927	1735	1729
	Std	6333	12,261	7606	6930
Employment	*Mean*	2630	6698	3330	3416
	Std	10,764	23,293	12,442	10,584
	N	4440	8769	1406	1443
Share of formal firms	*Mean*	0.371	0.359	0.328	0.322
	Std	0.305	0.242	0.275	0.251
Share of formal employment	*Mean*	0.413	0.434	0.377	0.386
	Std	0.315	0.272	0.289	0.278
	N	4060	8418	1304	1359

Note: Core sample contains 38 backward districts with gradation scores between 351 and 500 and 39 non-backward districts between 501 and 650. The light manufacturing contains seven 2-digit industries, heavy manufacturing nine industries, and ineligible industries 36 industries plus 1 excluded category. For reporting formalization rates, the industry-by-district cells without any firm are dropped.

Table 9 presents means and standard deviations of 2005 data. Compared to 1998, we see significant growth in the number of firms and employment in light manufacturing and non-manufacturing industries between 1998 and 2005. However, the gaps in number of firms and employment across districts remained substantial and even larger. For instance, the number of light manufacturing firms increased by 48% (=1415/953) in backward districts and 202% (=2438/1208) in non-backward districts. When we focus on the core sample of districts, where backward districts had much higher means of firm count and employment than non-backward districts in 1998, we find the means to be almost equal between districts in 2005. On the other hand, the average

[21] We note that Chaurey (2017) does not find any spatial spillovers in his evaluation of India's New Industrial Policy (NIP). To reconcile these different findings, we argue that spatial displacement is more likely to take place between the treated and untreated areas, which are fairly close in other aspects. Himachal Pradesh, one of the two states covered by NIP, was designated as backward state in 1993. In contrast, Punjab, its neighboring state, was considered an advanced state in the country. All 12 districts of Punjab had very high gradation scores. It would be very hard to attract firms from Punjab to Himachal Pradesh with limited policy benefits.

[22] We refrain from inferring the overall program effects for the whole country as it requires estimating direct effects and spatial spillovers on all districts and converting them from percentage changes to size changes of each district-industry. The policy setting we exploit here only enables us to obtain credible estimates about a marginal shift near the cutoff. We thank one of our referees for pointing this out.

Table 10
Program impacts in 2005.

VARIABLES	Number of firms			Employment		
	(1)	(2)	(3)	(4)	(5)	(6)
Backward district	0.037			−0.004		
	(0.121)			(0.126)		
Backward district * eligible manufacturing		0.133			0.133	
		(0.138)			(0.146)	
Backward district * eligible light manufacturing			0.194			0.199
			(0.145)			(0.154)
Backward districts * eligible heavy manufacturing			0.086			0.082
			(0.160)			(0.171)
Backward district * ineligible industries		−0.005	−0.005		−0.063	−0.063
		(0.123)	(0.123)		(0.127)	(0.127)
Observations	4081	4081	4081	4081	4081	4081
R-squared	0.837	0.837	0.837	0.817	0.818	0.818
	Share of formal firms			Share of employment in formal firms		
	(7)	(8)	(9)	(10)	(11)	(12)
Backward district	0.080*			0.063		
	(0.045)			(0.044)		
Backward district * eligible manufacturing		0.088*			0.080	
		(0.047)			(0.048)	
Backward district * eligible light manufacturing			0.096**			0.086*
			(0.048)			(0.049)
Backward districts * eligible heavy manufacturing			0.081*			0.076
			(0.048)			(0.051)
Backward district * ineligible industries		0.077*	0.077*		0.056	0.056
		(0.045)	(0.045)		(0.044)	(0.044)
Observations	3814	3814	3814	3814	3814	3814
R-squared	0.403	0.403	0.403	0.357	0.358	0.358
State dummy	Y	Y	Y	Y	Y	Y
2-digit industry dummy	Y	Y	Y	Y	Y	Y
2nd order polynomials of gradation score	Y	Y	Y	Y	Y	Y
1991 district covariates	Y	Y	Y	Y	Y	Y

Notes: The core sample including backward districts with gradation scores ranging from 351 to 650 is used in the regressions. The industry-by-district cells with zero firms are excluded for regressions of formalization rates. Gradation score is normalized by dividing the original score with 500 and then subtracting 1. Standard errors in parentheses are clustered at district level. *** indicates significance at 1%, ** significance at 5%, * significance at 10%.

formalization rates of light manufacturing in backward districts exceed those in non-backward districts with a modest margin of about 2%.

We estimate equations (1)–(3) with EC 2005 data for our core sample using all four dependent variables. The results are reported in Table 10. The coefficient estimates for number of firms and employment decline to 19% for light manufacturing and are statistically insignificant. Estimates also drop to about 8% for heavy manufacturing and turn negative for ineligible industries (both statistically insignificant). The numbers clearly suggest that the gains in manufacturing productions from the program did not last long after it concluded.

One plausible explanation is that the program stimulated manufacturing sector in the better-off backward districts mainly by encouraging firm entry rather than firm growth. Once the program stopped, not only did new firms lose incentives to enter these districts, but factors such as market access and infrastructure conditions would work to restore the initial spatial equilibrium. In the literature that identifies persistent effects of place-based policies (Ehrlich and Seidel, 2018; Kline and Moretti, 2014), improved infrastructure is recognized as a critical factor for the impact of a policy to continue after it ends. In the case of the backward district program, however, the main intervention was a tax exemption offered to firms. Thus, the government was unlikely to have enough public resources to invest in additional local infrastructure, which is necessary for sustaining agglomeration of manufacturing. Therefore, we see the program's effects dissipate quickly.

For formalization rates, the results show that the program increased the share of formal light manufacturing firms or their employment in backward districts by about 9% with statistical significance at 5% or 10% level. The increases for heavy manufacturing and ineligible industries were 8% for the number of firms and 6–8% for employment, though the estimates are less precise. This is interesting as it suggests that the program's effect on formalization become more pronounced in the long run than in 1998 when the program was still in place.[23] It seems unlikely that this effect was driven by the arrival of more registered firms in backward districts since the effect of the program on number of firms diminished by 2005. More plausibly, originally unregistered firms may have obtained official registration status as a response to the program. While more research is needed to understand why the program's effect on formalization emerged in the longer term, it is possible that the tax exemption induced small increments in firm registration in the short run, which had a large, cumulative impact on local business culture by demonstrating that for some firms at least, formalization could bring benefits (Fajnzylber et al., 2011).

7. Conclusion

Place-based policies have been popular in both developed and developing countries. We evaluate the backward district program initiated by the Indian government in 1994, which aimed to promote manufacturing in underdeveloped areas across the country. The way the program selected backward districts allows us to reliably estimate the program's impacts on industrial development, at least for a subset of treated districts.

We find that over the short run the program led to faster growth in light manufacturing in the better-off backward districts. The substantial positive impact was unlikely to have been driven by the possibility that more firms became registered in response to the program's tax exemption and thus got included in the economic census. Instead, our analysis

[23] The results should be taken with a grain of salt since, as mentioned earlier, registration rates may be underestimated as EC 1998 missed a few important categories of registration.

suggests that part of the effects could be attributed to spatial displacement occurring across neighboring non-backward districts.

Further, the program's impacts diminished greatly over time, implying that the program failed to generate self-sustained growth in the target industries in treated areas. The finding, though disappointing, bears important implications for the design of future place-based development policies. First, an array of constraints to business are likely to exist in underdeveloped areas, such as the absence of good transport and energy infrastructure and urban towns with amenities to attract skilled workers and managers. Tax benefits alone are unlikely to counter the full drag of underdevelopment. A multifaceted approach is thus more desirable for underdeveloped areas. In this regard, other types of place-based policies such as special economic zones or industrial parks that provide a comprehensive package to potential investors, including better infrastructure services, the opportunity to co-locate with other firms (some of whom may be buyers or suppliers), and more streamlined business processes (such as single window clearances) may be more effective to address development constraints. In this context, it is worth highlighting that infrastructure is a key input to industrial activity and was generally in short supply to underdeveloped regions. Given its longevity, public investment in infrastructure could be more cost effective than pure tax incentives in promoting agglomeration externalities.

Finally, for countries like India that have relatively low internal migration rates (Munshi and Rosenzweig 2016), industrialization may well progress faster with policies that facilitate the movement of labor to places with existing agglomeration or comparative advantages in manufacturing (for example, by providing more affordable housing) than policies trying to take manufacturing to the disadvantaged regions.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jdeveco.2020.102621.

Table A1
Indicators used to construct gradation scores to identify backward districts

Criteria	Weights	Indicators
Financial	3	Per capita credit given by scheduled commercial banks
	2	Per capita deposit received by scheduled commercial banks
Infrastructural	1	Phones per thousand population
	2	Per capita power consumption
	1	Urbanisation (urban population of a district as a proportion of total population)
	1	Metaled roads per 100 square kilometers
Industrial	3	Workers in registered factories per thousand population (excluding electrical undertakings and bidi & cigar units)
	2	Per capita gross value added from registered manufacturing sector

Source: "All India Gradation List", Appendix III of the Income Tax Act. Notification of Government of India, Ministry of Finance, Department of Revenue (Central Board Direct Taxes) S.O. 635 (E); Accessed at http://ncrpb.nic.in/pdf_files/16_AnnexureVII_part1_cma.pdf.

Table A2
Program impacts on number of firms and employment in 1998: expanded samples

	Number of firms			Employment		
VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
Full sample						
Backward district	−0.0696*			−0.121***		
	(0.0376)			(0.0457)		
Backward district * eligible manufacturing		−0.136**			−0.473***	
		(0.0570)			(0.0733)	
Backward district * eligible light manufacturing			0.213***			0.0119
			(0.0715)			(0.0857)
Backward districts * eligible heavy manufacturing			−0.408***			−0.849***
			(0.0672)			(0.0878)
Backward district * ineligible industries		−0.0496	−0.0496		−0.0152	−0.0152
		(0.0386)	(0.0386)		(0.0473)	(0.0473)
Observations	24,564	24,564	24,564	24,564	24,564	24,564
R-squared	0.828	0.828	0.828	0.795	0.796	0.797
Restricted sample (gradation scores between 251–850)						
Backward district	0.00261			0.0376		
	(0.0650)			(0.0751)		
Backward district * eligible manufacturing		0.126			0.0299	
		(0.0824)			(0.102)	
Backward district * eligible light manufacturing			0.290***			0.261**
			(0.106)			(0.126)
Backward districts * eligible heavy manufacturing			−0.00131			−0.150
			(0.0931)			(0.115)
Backward district * ineligible industries		−0.0347	−0.0347		0.0399	0.0399
		(0.0684)	(0.0684)		(0.0790)	(0.0791)
Observations	11,178	11,178	11,178	11,178	11,178	11,178
R-squared	0.827	0.827	0.827	0.796	0.796	0.796
State dummy	Y	Y	Y	Y	Y	Y

(continued on next column)

Table A2 (*continued*)

	Number of firms			Employment		
VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
2-digit industry dummy	Y	Y	Y	Y	Y	Y
2nd order polynomials of gradation score	Y	Y	Y	Y	Y	Y
1991 district covariates	Y	Y	Y	Y	Y	Y

Notes: The restricted sample includes backward districts with gradation scores ranging from 251 to 850. The dependent variable was transformed as log(Y+1). Gradation score is normalized by dividing the original score with 500 and then subtracting 1. Standard errors in parentheses are clustered at district level. *** indicates significance at 1%, ** significance at 5%, * significance at 10%.

Statement

The data used in the paper titled "Place-based Preferential Tax Policy and Industrial Development: Evidence from India's Program on Industrially Backward Districts" include Economic Census 1998, Economic Census 2005, Primary Census Abstract 1991 and districts' gradation scores published in the "All India Gradation List" of the Income Tax Act. The raw data are all publicly available. The processed data are available from authors upon request.

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